

Assessment 2

GroupProject

Course: INFS2603 – Business Analysis and Agile Product Management

Group Number: W18A Group 5

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Business Need

Business Need Identification

Coditas finds itself under increasing pressure to scale its business processes and delivery high quality solutions for its client at the same time. The pressure stems from rising demand, changing technology and growing competitive pressures within the digital engineering industry. Being a knowledge-intensive organization, Coditas needs to utilize its engineering talents to meet the satisfaction from their clients. However, the skills of their engineers are only limited within project managers' memory and not effectively captured, leading to inefficient deployment of talents. Therefore, Coditas requires the capability to make their engineers skill visible to the whole organization and usable for effective staffing decisions to reduce the project duration with better quality.

High-Level Business Requirements

- Capture and maintain up-to-date information on engineers' skills and experience.
- Search and identify suitable engineers based on project requirements.
- Support managers in making informed staffing decisions.

Desired Outcomes

- Faster and more efficient team formation.
- Improved utilisation and allocation of workforce talents.
- Enhanced scalability of staffing decisions as the organisation grows

Analysing the context

The business need of Coditas is affected by factors from both the external business environment as well as organizational constraints from inside, which put pressure on Coditas to enhance its management of workforce capabilities.

External Environment

Skill Shortage

Coditas operates within the Australian computer system design services industry, which currently faces a high demand for its products and services. However, the industry is currently experiencing issues to meet this demand in the future due to insufficient workforce capacity. Under the current supply, the industry research highlights that Australia still needs over 100,000 additional IT professionals by 2027 (Campbell 2025). This indicates that

organisations can no longer solely depend on hiring to solve the skill gaps but also improve how they leverage their existed internal skills to sustain delivery performance.

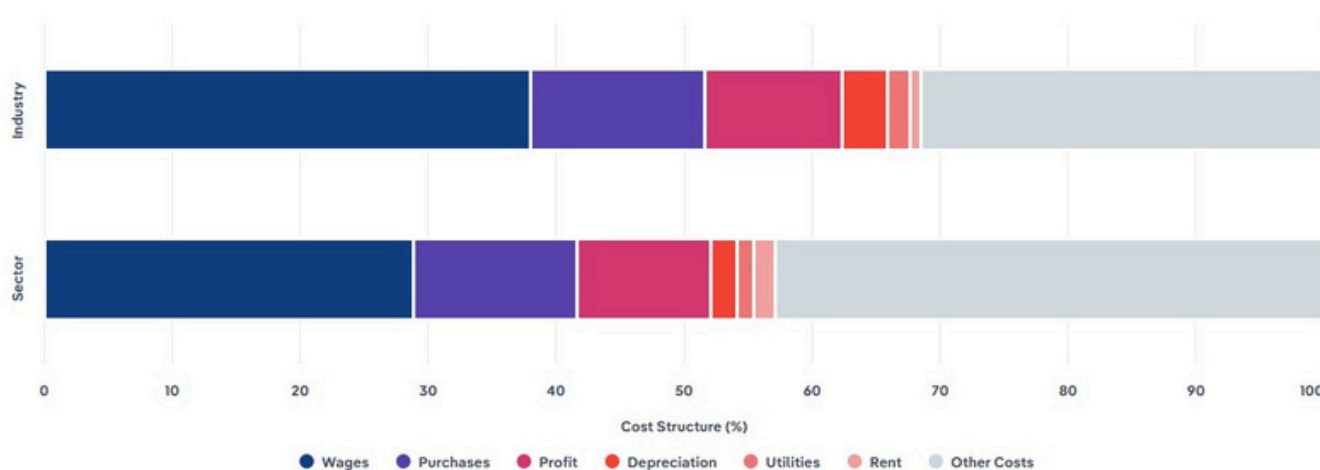
Competition Pressure

On the other hand, Coditas operates in a highly competitive industry environment with both onshore and offshore service providers. Offshore firms benefit from significant cost advantages, which leads to lowering labour costs by 50-70% (Putri 2025). At the same time, the onshore providers distinguish themselves by their close collaboration with the clients and higher local expertise. As cost advantages are widely shared and competition is high, firms cannot rely on pricing to differentiate. Instead, the competitive advantage for a company lies in its efficiency in execution capabilities and high-quality services. Ultimately, the delivery of high-quality services lies in the capability of the engineers within the organization, making effective talent management critical for sustaining this competitive advantage.

Human Capital as a Key Success Factor

Computer System Design Services in Australia Cost Structure Benchmarks

Average operating costs by industry and sector as a share (%) of revenue 2025



Additionally, the industry is highly dependent on human capital, with labour making up to approximately 40% of the company cost structure. Engineers are both the largest cost item and the main contributor to the company's revenue, with each generating 300,000 AUD annually (Campbell 2025). Hence, inefficient utilisation of workforce capabilities not only impact profitability and operational performance but also risk of undermine the competitive position in the industry. As such, the mismanagement of labour skills results not only in reduced efficiency and profits but also weakening the company's competitive position.

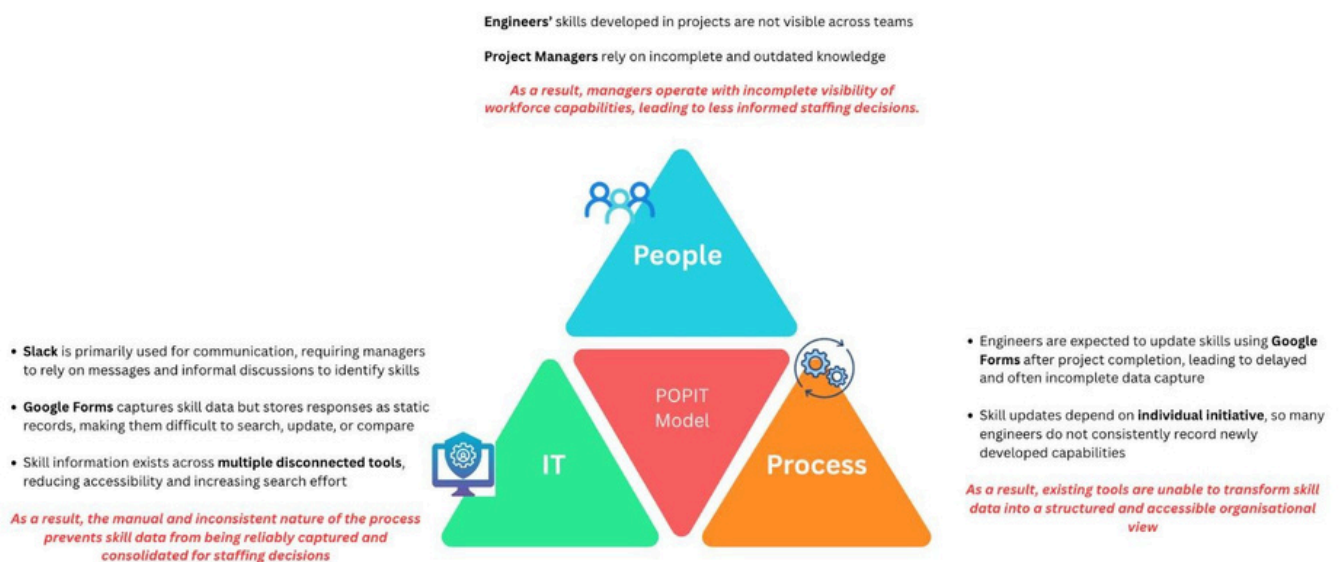
Implication for Coditas

These factors together underscore the necessity for Coditas to optimize its management of employee skills. Skill shortage restricts the ability for potential expansion via recruitment, while competition demands quicker and more precise creation of teams capable of producing better results for clients. Additionally, the economic importance of human capital increases the cost of inefficiencies and waste of talent utilisation. Therefore, improving the visibility and accessibility of workforce skills is not only operational improvement but a necessary capability for Coditas to remain competitive, utilise resources efficiently to drive sustain growth.

Internal Context

POPIT Analysis

To understand the internal factors contributing to limited skills visibility, Coditas' operating environment is analyzed using the POPIT model.



Overall, the skill data in Coditas is incomplete, inconsistent for organizational purposes. Although many engineers record their skills through tools such as Google Forms, the updates cannot be captured properly as everything depends on manual entries. Meanwhile, the recorded data is scattered throughout various places and fails to produce a coherent picture. Consequently, the managers fail to see a comprehensive view of all available skills and end up making inefficient and suboptimal staffing decisions.

Stakeholder Analysis

1. Stakeholder Identification and Categorisation

To understand the organisational challenge of limited visibility of evolving skills, key stakeholders were identified and categorised based on their level of involvement with staffing, skill development, and decision-making processes.

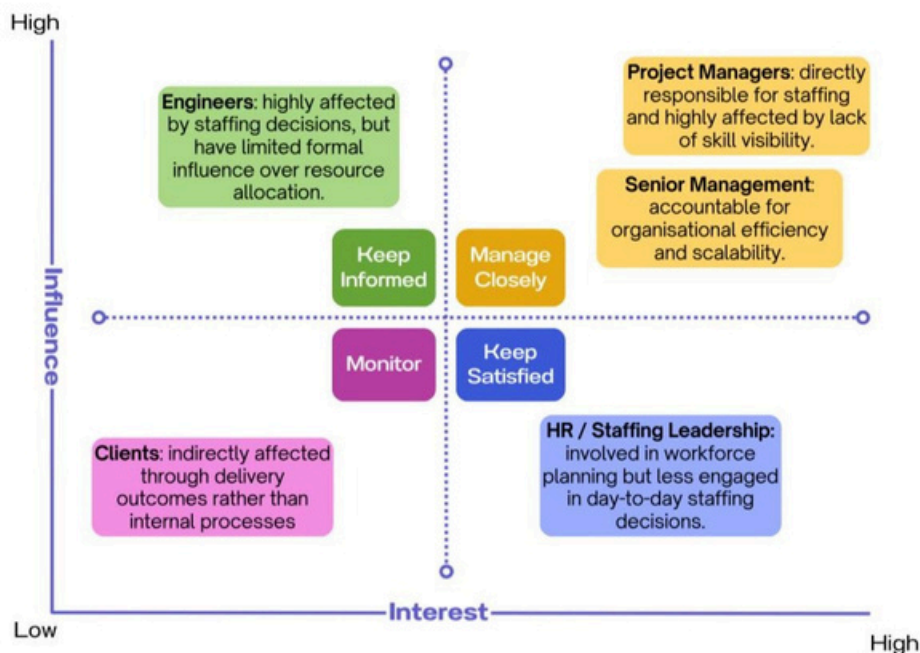
Primary stakeholders are those who directly interact with or are impacted by the proposed solution:

- Engineers: responsible for delivering project work and continuously developing technical skills.
- Project Managers (PMs): responsible for allocating staff directly working with projects and ensuring timely delivery.

Secondary stakeholders are indirectly impacted but also play an important role in delivering outcomes:

- HR / Staffing Team: supports workforce planning, hiring, and capability development.
- Senior Management: responsible for strategic oversight, utilisation rates, and organisational performance.
- Clients (indirect): affected by staffing quality and delivery efficiency.

2. Influence-Interest Matrix



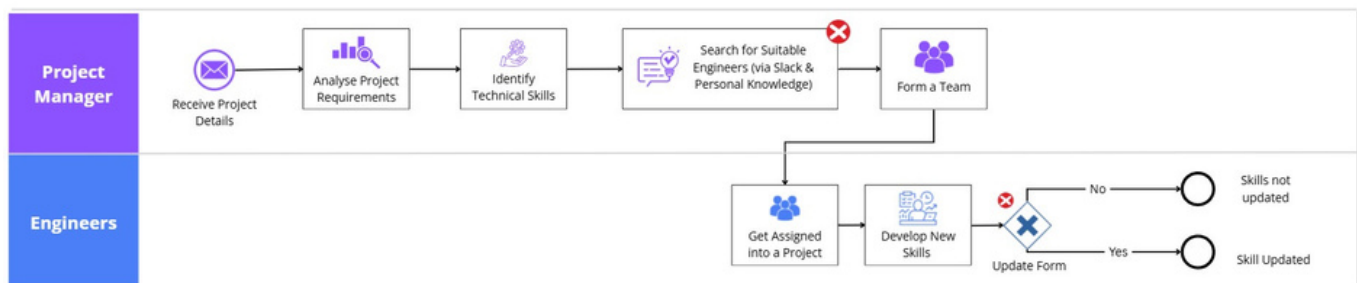
The matrix highlights a structural imbalance where decision-making authority is concentrated among high-power stakeholders (project managers and senior management), while engineers, despite being key contributors of skill data, have limited influence in making decisions for their colleagues. This reinforces the organisation's reliance on informal knowledge rather than structured, data-driven decision-making.

3. Stakeholder Needs and Pain Points

A deeper analysis was conducted to identify each stakeholder's goals, challenges, and requirements in relation to the problem.

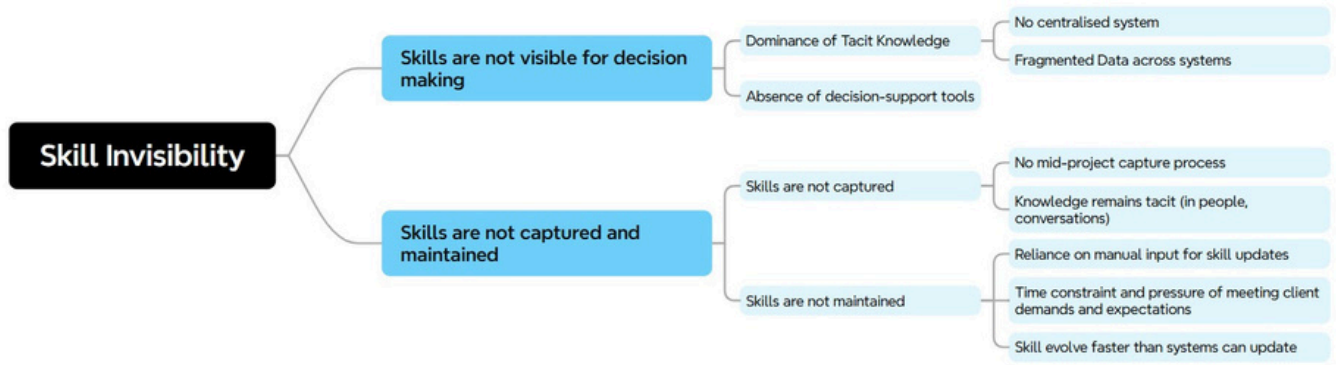
Stakeholders	Needs / Goals	Pain Point
Engineers	Be recognised and assigned based on their current skills	Skills not consistently recognised or updated
Project Manager	Allocate right people into projects quickly and accurately Gain insight into organisational skills for	Lack of reliable data of staffing skills
Senior Management	strategic planning Maintain a clear and consolidated view of	Limited visibility of skill distribution across the organisation
HR	workforce capabilities	Inconsistent data of workforce
Clients	Recieve high-quality delivery or outcomes	Variability in outcomes due to suboptimal staffing

Root Cause Analysis



The AS-IS model highlights two critical breakdown points: identification of suitable engineering during staffing and the updating of skills during project execution. These breakdowns reveal that skills invisibility at Coditas is not an isolated issue, but a systemic problem arising from failures in how skills are captured, maintained to support for decision-making.

To analyse this, a root cause tree was developed, decomposing two key dimensions: skills are not visible for decision making and skills are not captured and maintained. Together, these dimensions reflect the lifecycle of skills within organisations, from creation to utilisation.



1. Skills are not visible for decision making

The BPMN shows that managers rely on informal channels such as Slack and personal knowledge to identify suitable engineers, indicating that existing skills are not visible in a structured or accessible form. This reflects a deeper issue where organisational knowledge exists but is not surfaced in a way that supports decision-making.

A key driver of this problem is the dominance of tacit knowledge within the organization. Skills developed through project are often embedded within individuals and informal interactions, rather than being codified into structured systems. Research highlights that tacit knowledge plays a critical role in in problem-solving and innovation but is inherently difficult to access and share without appropriate mechanisms to make it explicit (Conneally 2025).

To contribute to the difficulty of accessibility of tacit knowledge is that they spread across different communication tools. While information about engineers' capabilities may exist, it is not aggregated or presented in a decision-ready format. As a result, managers rely on memory and informal networks, limiting their visibility to a small subset of individuals. Finally, without a tool that supports the manager to quickly search, filter and matching of skills to project requirements, existing capabilities cannot be effectively utilised.

2. Skills are not captured and maintained

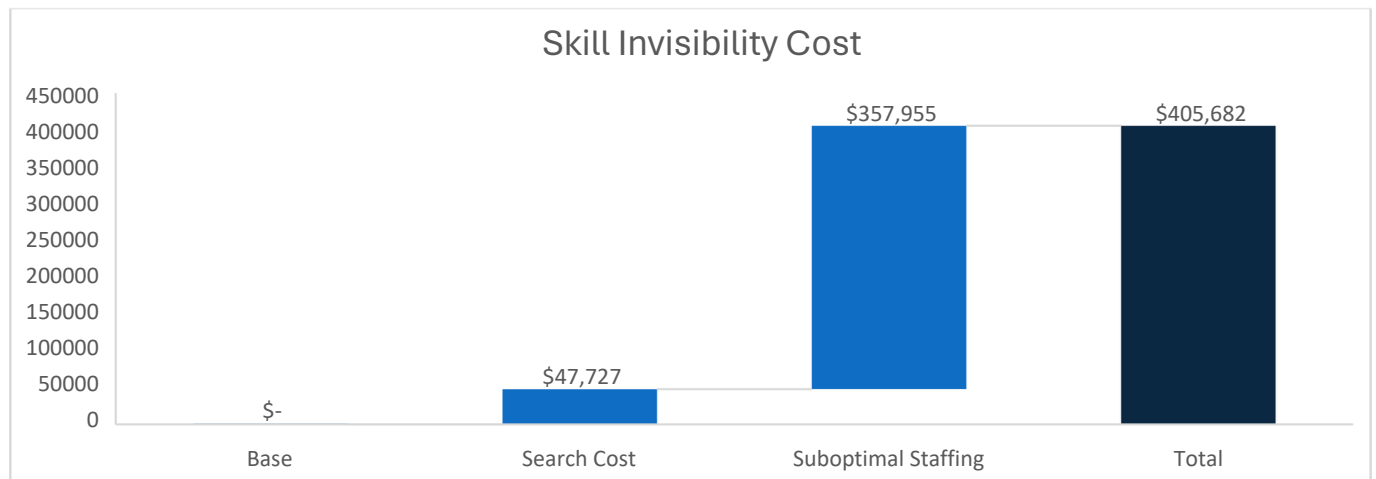
First, this happens because skills are not captured during project execution. Coditas confirms that documentation primarily occurs at the beginning and end of projects, leaving a critical gap during execution where most learning takes place. As a result, newly acquired skills are not transformed into organizational knowledge and remain invisible beyond the immediate project context.

Second, skills are not effectively maintained, as the current process relies on manual input from engineers. The BPMN shows that engineers are required to update their skills through forms, but this step is often skipped.

Specifically, Coditas has attempted such approaches, but adoption was low due to time constraints and competing projects priorities. Research supports this behaviour, efficiency and value delivery got prioritised first, and non-essential tasks such as manual updates are often avoided in value stream management (Takakura n.d.).

Economic Impact

The identified root causes create a fragmented and unreliable view of workforce capabilities, limiting Coditas' ability to effectively identify and deploy internal talent. As a result, these issues manifest in operational inefficiencies that translate into measurable economic and strategic impacts.



1. Search Friction

Low skills visibility leads to inefficiency in terms of searching for potential candidates within the company. Hence, there are potential costs associated with the use of managerial time. According to the estimates, the cost equals to \$47,727 per year (Appendix A).

2. Suboptimal Staffing

Another negative implication is that Coditas is more likely to not hire the best possible engineers based on the requirements of their projects. In such situations, engineers may not have the proper skills and abilities to deliver a specific job effectively.

As a result, it will take longer for them to get started because they would have to learn how to use the required tools. It can also lead to lower quality due to higher operational errors (Vantighem 2026).

At the same time, some employees who would be better for particular jobs are less visible to other Coditas managers. This means that the organization may miss out on the utilization of those talents. Our analysis suggests that Coditas can potentially loses \$357,955 per year because of ineffective staffing (Appendix A).

3. Scalability Constraint

Finally, it should be noted that the inefficiency caused by low skills visibility can become more severe as the company scales. The increase in both the number of employees and the size of the company makes it more difficult to maintain efficient communication, leading to greater search cost and suboptimal staffing. The sensitivity analysis (Appendix

A) demonstrates that if the number of employees increases by 100 while increase in 5% suboptimal allocation takes place, economic costs will rise by 45%, totalling \$588,636.

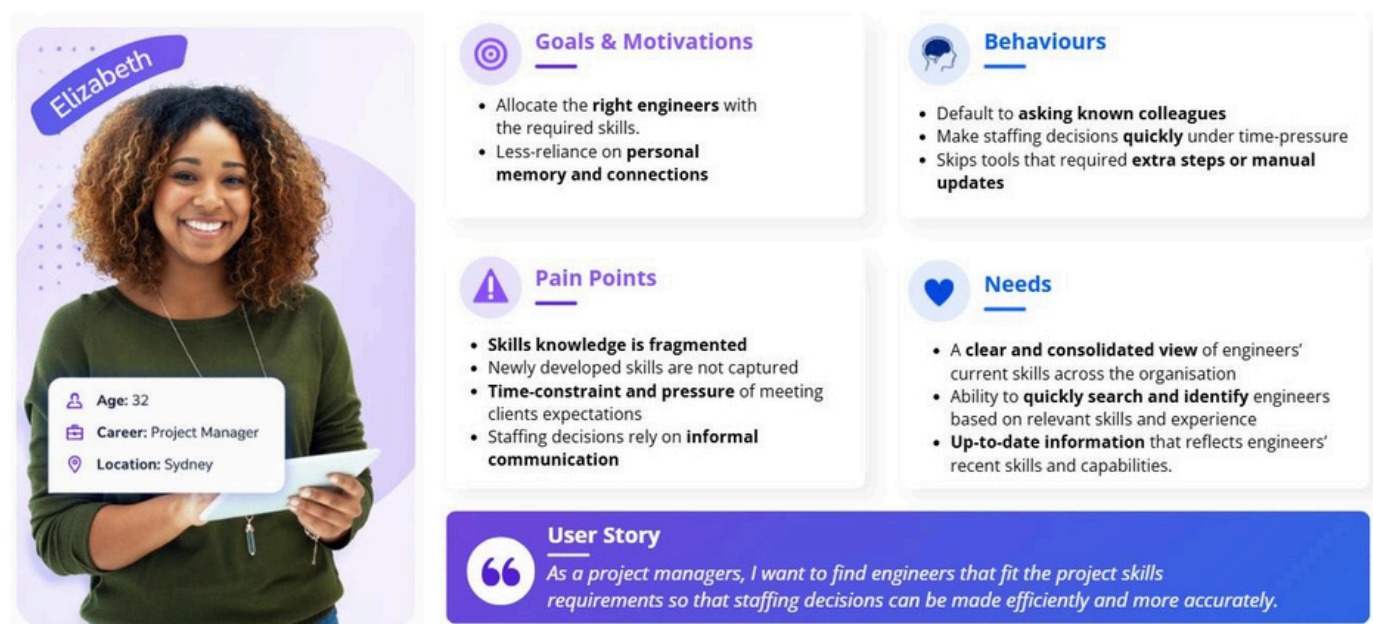
Problem Statement

Coditas lacks the capability to capture and surface engineers' evolving skills in a structured and accessible way. Consequently, project managers rely on informal knowledge and fragmented information to make staffing decisions, leading to inefficient search, suboptimal team allocation and reduced scalability. This is not due to lack of skills but the absence of system that makes those skills visible, reliable for decision making.

Product Vision

We help project managers to quickly assemble high-performing teams by making engineers' skills visible and accessible across the organization.

Persona



Elizabeth

Age: 32
Career: Project Manager
Location: Sydney

Goals & Motivations

- Allocate the **right engineers** with the required skills.
- Less-reliance on **personal memory and connections**

Behaviours

- Default to **asking known colleagues**
- Make staffing decisions **quickly** under time-pressure
- Skips tools that required **extra steps or manual updates**

Pain Points

- Skills knowledge is fragmented**
- Newly developed skills are not captured
- Time-constraint and pressure** of meeting clients expectations
- Staffing decisions rely on **informal communication**

Needs

- A **clear and consolidated view** of engineers' current skills across the organisation
- Ability to **quickly search and identify** engineers based on relevant skills and experience
- Up-to-date information** that reflects engineers' recent skills and capabilities.

User Story

As a project managers, I want to find engineers that fit the project skills requirements so that staffing decisions can be made efficiently and more accurately.

- **Name:** Elizabeth
- **Age:** 32
- **Status:** Project Manager
- **Value:** Seeking efficiency and accurately in staffing allocations and auto update data.

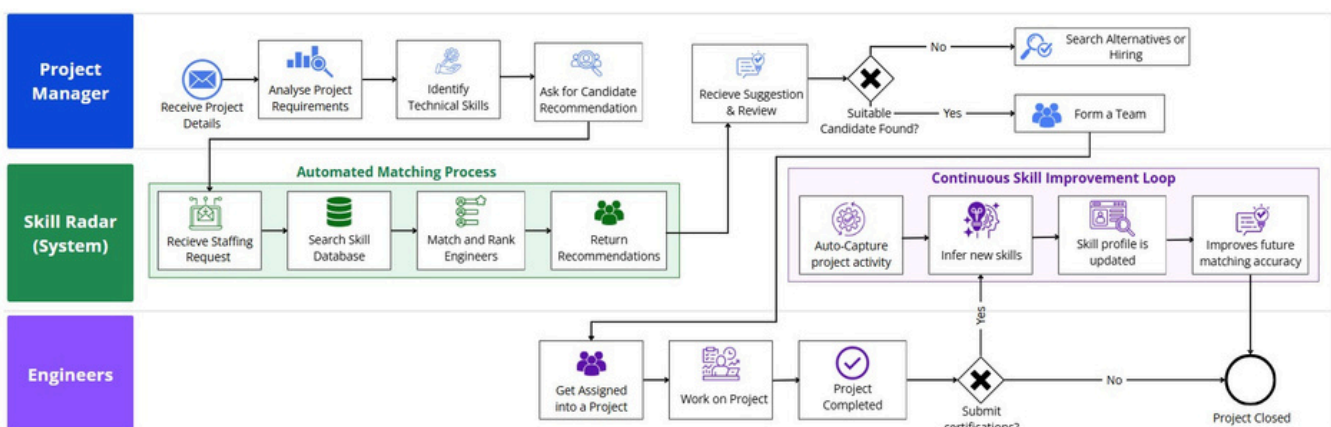
- **Problem Overview (Doing):** Despite her efforts in finding and searching for suitable engineers in each project, the skill data and related information is fragmented and lacked visibility.
- **Result (Feeling):**
 - She's continuously experiencing these inefficiencies, causing her to feel overwhelming and frustrated about is that engineers suitable for those projects.
 - Spending more time than usual for seeking and evaluating staff with inconsistency data in the whole systems make Elizabeth feel stressed and give a wrong choice if she overlooks the capable engineers or assign individuals whose skills do not fully match project requirements.

Solution Design

To address the problem that Coditas is facing, which is heavily rely on managers' personal knowledge when making staffing decisions and also data is fragmented based on manual update, a skills visibility and intelligent staffing system runs to help project managers to make recommendations based on real-time workforce capabilities. However, several assumptions were made to evaluate the effectiveness of the solution:

- Availability of reliable skills data: engineers' skills and project experiences can be captured with sufficient accuracy and confidence, through manual updates or inferred from project work.
- Standardised skill representation: consistent skill taxonomy can be established across the organization, allowing skills to be comparable, searchable and usable for matching
- Sufficient data for matching logic: there is an adequate data on skills and project experience to support meaningful recommendations for the AI system.

TO-BE Diagram



After implementing a new systems called Skill Radar, business process has improved activity-related and outcome-related delivery from the previous workflow.

1. Automate Matching Process

With this new system, the reliance on personal knowledge or manually finding required engineers will be eliminated. This optimises time quality and accuracy while searching for workers. The systems can search its database to returning recommendations with clear ranking, enabling project managers to efficiently identify the most suitable engineers. Once completed, the system will be automatically required a review and approval from managers.

2. Skills Data Improvement Loops

The activity of continuous skill improvement loop, which is occurring during the whole project when engineers commit their code on the workspace. This new function is to address the lack of visibility in capturing the workloads from employee during projects. Consequently, the system automatically captures project activities and infers new developed skills, then auto-update its database to improve accuracy for future decisions. Over time, this creates a self-reinforcing system where each project enhances the quality of workforce data and decision-making.

Key Improvements

Before	After
Skills not visible to managers	Skills clearly visible the system
Reliance on tacit knowledge	Structured, data-driven skill profiles
Manual search for candidates	Automated matching & ranking system
Suboptimal staffing decision	Best-fit team selection based on project requirements
Static, outdated skill capture	Dynamic, evolving capability visibility

Feasibility

1. Technical Feasibility

The proposed solution is technically feasible given Coditas' existing environment of digital engineering capabilities and agile teams.

- Functional risk: Coditas teams are already familiar with project delivery workflows, but not with structured knowledge capture, however, Skill Radar introduces adoption friction rather than capability limitations.
- Technology risk: the solution can leverage existing tools (e.g., Slack, Git, etc.) for automated knowledge capture, which reduces heavy reliance on manual input critical given prior failure of form-based systems.

Project size risk: the new systems only or recommended to be used for a small group of employees in Coditas such as management teams, HR department to avoid integration across multiple teams and increases complexity.

Compatibility risk: integration with existing tools (Slack, Jira, GitHub) is essential, if not be able to integrate

- will reduce adoption and effectiveness.

2. Organisational Feasibility

Strategic alignment: the solution directly addresses key organisational challenges which bias knowledge memory, skills invisibility, onboarding delays, and uncaptured the poorly maintain skill database.

Management support: these challenges impact scalability and delivery efficiency, which are critical at Coditas' growth stage.

3. Economic Feasibility

The proposed solution is significantly improved Coditas revenue and sunk costs and provide healthy financial impact of current inefficiencies in Coditas' workforce.

- Cost-benefit alignment: existing inefficiencies cost approximately \$405,682 annually, primarily from search friction and suboptimal staffing. The reduction of these losses from 20-30 percent will directly generates substantial recoverable value and saving from the operational costs.
- Value recovery potential: improved skill visibility directly reduces managerial search time and misallocation of engineers, translating into higher utilisation rates and faster project delivery directly improving financial health.
- Scalability impact: projected costs rising to \$588,636 due to inefficiencies scale with organisation size can be conducted with the new system and prevent further financial losses.
- Cost structure: Skill Radar leverages existing infrastructure which minimising upfront development and large investment costs, which easier to gain a positive outcome in short-term run.

User Story Mapping

	Define Project	Generate Candidates	Evaluate Candidates	Build Team	Skill maintenance
User tasks	Define project requirements (skills, roles, constraints) Input Project Details (Domain, Type, Description)	View the list of recommendations Read the AI recommendation justifications	View the candidate profile in details Understand candidate fit Compare candidates	View assembled team View skill coverage Add/remove candidates	Review skill updates Confirm/update skills
Release 1	Create project request Input skills & Constraints	AI candidates recommendation systems Candidates ranking weighting Candidate Selection Justification	Candidate Profile Page Skill and experience database	Select engineers based on system recommendation View skill & experience coverage	Integrate past experiences into new system database Automatic notifications for skill capture & updating
Release 2	Upload project brief to automatically fill the form	Adjust ranking based on filters	Skill taxonomy	Sample team generation	

The product backlog was purposely developed to translate the product vision into a structured deliverable; this prioritises value creation for project management process while addressing the core problem of skills invisibility. The backlog was split into 2 delivery phases base on the business value, dependency, and contribution to reduce search friction and optimising staff allocation, moreover, to provide a systematic continuous improvement capability.

The first phase (Release 1) focuses on delivering the Minimum Viable Product (MVP), at this stage, the system will have the capability to enabling end-to-end staffing workflow allocation, including marking, ranking, and evaluating candidates, defining project requirements, then therefore, generating and ranking candidates' recommendations, team formation with clear visibility of skill coverage. These features directly address the core inefficiencies by provide additional support and key information that aid managers in team formation progress.

The second phase (Release 2) introduces advanced features aim at improving manager's decision accuracy, data reliability, and overall system usability. This includes filtering mechanisms, structured skill representations, and standardized skill taxonomy. In addition, supporting features such as skill inputs validation and evaluation, user flow and interface, as well as other micro-adjustments and automations will be continuously developed and optimized aim to deliver a more reliable system and evaluation accuracy, moreover, seamless experience and ease of use for all users.

Risks & Mitigation

1. System Unreliable

System reliability remains a big risk factor for the project to happen in the system during the early stages. Especially when the system does not have sufficient data points may result in managers will lean towards cognitive judgements instead of actual valuated numbers (Lincoln Laboratory MIT, 2026). Since this current project is identical to artificial

intelligence and machine learning processes, therefore, it will take time for the system to gain and train with a large data bank to become more accurate.

2. Data Incompleteness

Another crucial risk related to the inaccurate or incomplete data sets, which could reduce the quality of team formation and could introduce system bluffing (Mohammed et al., 2025). In order to mitigate this issue, the system will require continuous data training. Moreover, data that is being used to train must also be validated to remove outliers and reduce noise. Feedback loops, and user manual inputs and adjustments can also be considered as valuable data points that should be collected and analysed.

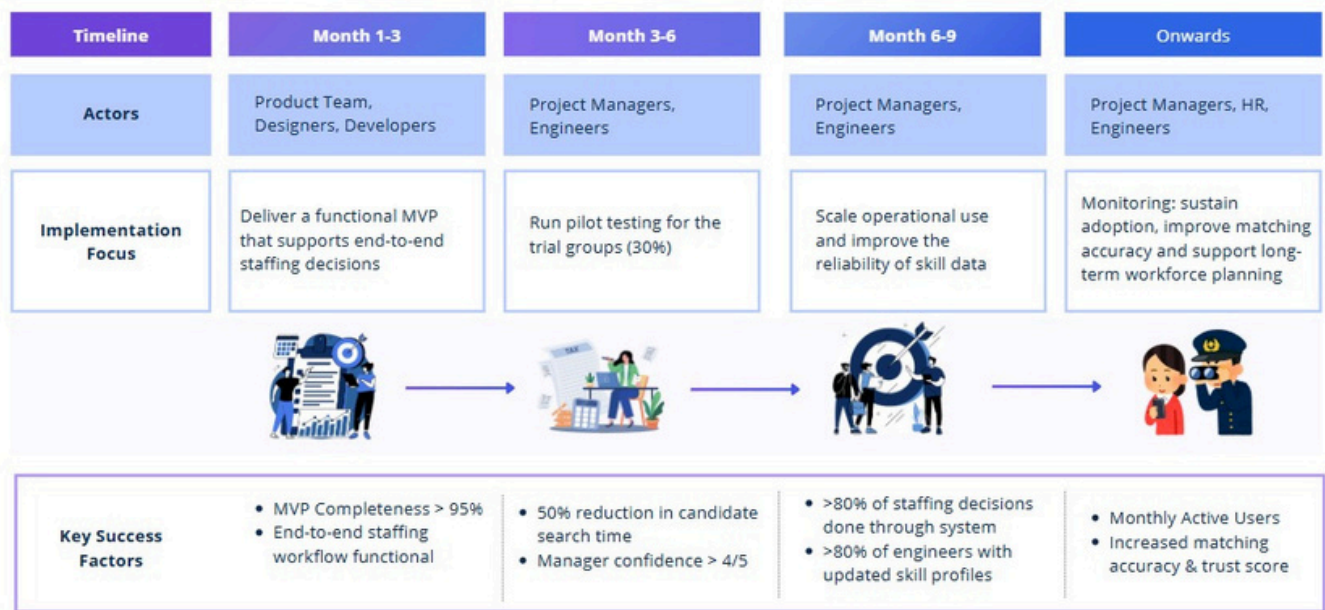
3. System Tailoring and User Adoption

Overfit and underfit are 2 critical problems for any machine learning system or learning models (LMs) (Miedema, Waschull and Emmanouilidis, 2026), the process of criteria building and training must be transparent and closely monitored, to therefore, improve the likelihood of user adoption by precisely deliver consistent, trustworthy and high- quality recommendations. Several pilots must be tested prior public launch to ensure a robust data set and consistent system recommendation; therefore, it is crucial to collaborate with large enterprises to have access and collect data points from the large pool of employees in early piloting stages. Moreover, the system will have to be designed to adapt to user/company behaviours, learn from the tasks and interactions from employees and companies, therefore, have a mechanism to adapt and tailor itself to that specific enterprise/company.

4. Ethical and Bias Risks

Algorithmic bias could pose a significant risk since skill inference and team matching heavily rely on historical, pre-analysed data, therefore, any preferential staffing patterns or teamwork capabilities between different individuals when grouped would require human inputs when needed. Moreover, the unequal in opportunities distributions will also need human inputs and reviews to avoid unfairness at workplace (Corrêa, Garsia and Elbi, 2025). On the system side, more tailoring micro-adjustments in algorithms will need to be thoughtfully designed to balance out performance over human cultural and social aspects during team-matching process including roles, demographics, races, opportunities, capabilities, etc.

Execution Roadmap & Product Analytics



The proposed solution will be implemented through a phased execution plan that allows Coditas to deliver value incrementally while validating whether the product improves staffing decisions.

In months 1-3, for focusing on the core workflow including project request capture, candidate matching, candidate profile viewing and team assembly, the team will work on the MVP with the completeness more than 95%. This must be the prioritisation for these functions due to its directly solving the Coditas's challenges. For the successful at this early stage, ensuring the tight scoped of MVP and workflow supports the user goal with productive team.

In the next 3-6 months, the MVP will be tested with a small group of project managers and engineers. The purpose of the testing is to evaluate whether the product is achieving its product goal: helping project managers quickly assemble high performing teams by making engineers' skills visible and accessible across the organization. The testing also validates several key assumptions underlying the solution. Specifically, whether the AI recommendation system can accurately identify the right candidates for specific projects with the availability and data quality that Coditas has.

To test this, participants will complete a scenario-based staffing task using the low-fidelity prototype. To measure success, the pilot group will be measured using the HEART framework, particularly Task Success, Engagement, and Adoption, as these are the most relevant categories for an internal decision-support tool.

For Task Success, the goal is to reduce the pain point of slow and uncertain staffing decisions. Signals of success include project managers completing staffing tasks more efficiently with greater confidence. Relevant metrics include time taken to identify suitable candidates, and manager confidence score after task completion.

For Engagement, the goal is reducing reliance on Slack, memory, and fragmented documents. Signals of success include active use of search, candidate profiles and team assembly features. Relevant metrics include average number of candidates searches per session, feature usage frequency per project cycle.

For Adoption, the goal is to reduce the manual input, a pain point for both project managers and engineers. Signals of success include choosing the system instead of informal methods and engineers maintaining usable profiles. Relevant metrics include percentage of staffing tasks initiated through the system, and percentage of engineers with updated skill profiles.

Continuously, the system can be rolled out more to the entire organization while improving data quality through structured skill updating and standardized taxonomy. Onwards, Coditas can add dashboard insights, and monitor the usage of systems to keep it updated to evolving skill development.

Project Management System

Our reflection in this section was centred on the way our group used the Agile Scrum methodology in practice, especially the way we executed the roles and ceremonies and how well they helped us to complete our task.

1. Scrum Roles

First, Scrum roles were determined through linking the duties of each role to the capabilities of individual team members, including their experience in working with Agile methods, as well as facilitative and analytical skills. The Product Owner (Gia Bao) was accountable for controlling the backlog and making sure that crucial parts such as problem definition and product vision stayed consistent with each other, thus dictating the general direction of the project. The Scrum Master (Bao Ngoc) was responsible for organizing discussions, coordinating activities and making sure that tasks moved forward as planned, especially during check-ins and reviews. The rest of the teams (Braynton, Aina, Fenghao) were the developers, responsible for researching, developing the MVP.

2. Sprint Planning

The top screenshot shows the Jira 'Backlog' view for the 'INFS2603_Group' space. The sprint is 'IG Sprint Week 7' (1 Apr - 7 Apr) with 12 work items. The items are listed with their IDs, descriptions, and status. The bottom screenshot shows the 'Timeline' view for the same sprint, displaying a Gantt chart with tasks and their durations. The tasks are listed on the left, and the timeline shows the duration of each task across the sprint weeks.

Work	26	27	28	29	30	31	1	2	3	4	5	6	7	8	9	10	11	12	13	14
IG-26 As a project manager, I want to receive a list of recommended engineers so that I ca...																				
IG-27 As a project manager, I want candidates to be ranked based on relevance so that I ca...																				
IG-29 As a project manager, I want to understand why candidates are recommended so tha...																				
IG-30 As a project manager, I want to view detailed profiles of enigneers so that I can evalu...																				
IG-31 As a project manager, I want o see engineers' skills, experience and past projects so ...																				
IG-36 As a project manager, I want to select engineers from system' recommendations so t...																				
IG-37 As a project manager, I want to view the overall skill coverage of the team so that I ca...																				
IG-28 As a project manager, I want to adjust filters (e.g. availability, experience level) so tha...																				
IG-32 As a project manager, I want to view engineers' skills in a structured format so that I ...																				
IG-34 As an engineer, I want to indicate my proficiency level for each skill so that managers...																				
IG-35 As a manager, I want skills to follow a standardised taxonomy so that information is c...																				
IG-33 As an engineer, I want to update my skills and experience so that my profile reflects ...																				

For sprint planning, Jira was applied to organize and prioritize the creation of product backlog. We organized the important activities such as the enterprise analysis, requirements and prototypes development as the items of backlog. For every Wednesdays, we will have a meetup after tutorial to choose which items in the backlog will be chosen for the next sprint. The priority and assignment of the activities were made accordingly to the work schedule and skills

of each member. Over time, we improved our backlog by adding new details and splitting tasks according to the problem defined.

3. Daily Scrum

For daily scrum, we made use of Whatsapp for communication about what had been achieved by each member and the tasks assigned to them in the form of three main points. These points were the tasks done, our next tasks and whether there were any impediments in completing these tasks. To achieve this, we used Jira alongside these updates to track task status where items were moved across stages such as “To-do”, “Doing”, “Review”, “Done”.

4. Sprint Review

During Sprint review, we held discussion sessions where each member presented their work and explained their reasoning with a focus on whether the analysis was logically developed and aligned with the project direction.

5. Retrospective

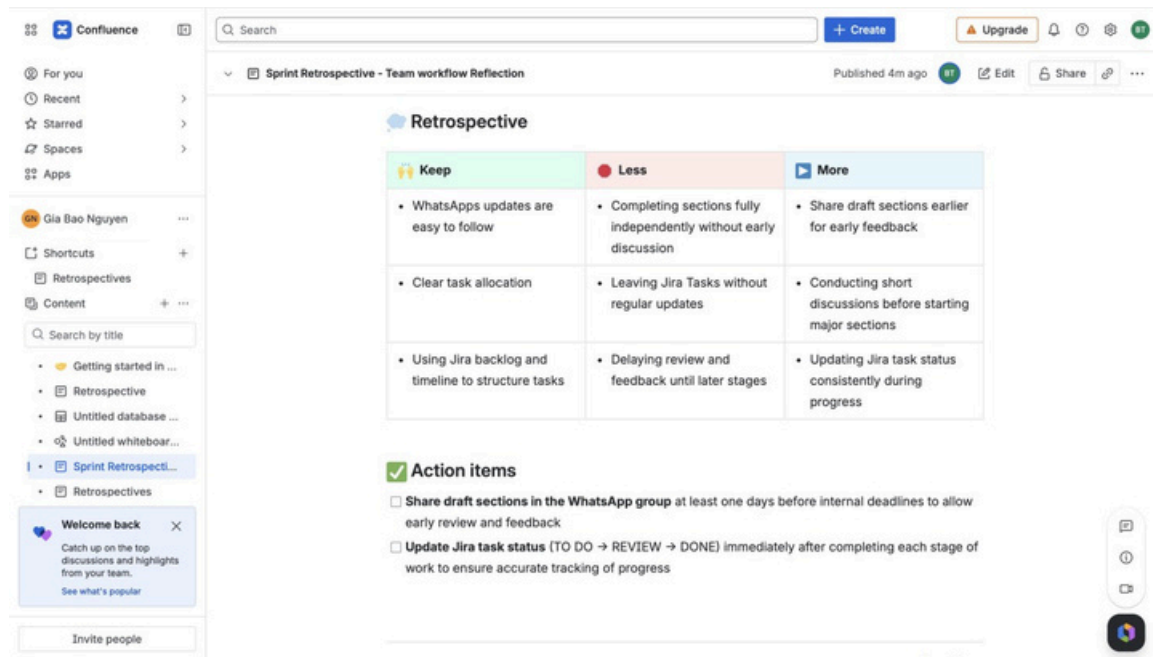
The screenshot shows a Confluence page titled "Sprint Retrospective - Team workflow Reflection" created by Brayton Huo Ylaw Tiong. The page includes a sidebar with navigation options like "For you", "Recent", "Starred", "Spaces", and "Apps". The main content area features a blue header with the title and a sub-header "Reflect on past workflow challenges and identify clear improvement opportunities for better collaboration". Below this is an "Overview" section with a table containing the following information:

Date	25 Mar 2026 Week 6
Team	INFS2603 - Group 5
Participants	@Brayton Huo Ylaw Tiong @Bao Ngoc Nguyen @Gia Bao Nguyen @Aina Madihah Mohd Nizam

Below the overview is a "Retrospective" section with three columns: "Keep", "Less", and "More".

Keep	Less	More
• WhatsApps updates are	• Completing sections fully	• Share draft sections earlier

The page also includes a "Welcome back" notification and an "Invite people" button at the bottom.



For retrospective practice, a structured retrospection board was utilized within Jira to conduct reflections on our workflow. We organized our ideas into categories such as “Keep”, “Less”, “More” which enabled us to articulate areas that went smoothly, areas that posed problems and those that required improvements. This approach made it easier for our members to express their opinions in a more organized manner enabling us to compare their differences and find out common issues. For instance, among the problems that were listed like “Delayed review and feedback”, the “Less” category, action plans such as sharing of draft sections in the WhatsApp group at least one day before the internal deadline were established.

Reference

Campbell, J 2025, *Computer System Design Services in Australia*, IBISWorld.

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Appendix

Assumption Table:

Items	Value	Source
Number of Engineers	750	Given
Cost per Engineer	\$100,000	IBISWorld
Projects per Year	35	Given
Working Days	220	Assumed
Days Wasted for Staff Allocation	3	Assumed
% Suboptimal Staffing	15%	Assumed
Inefficient Days	14	Assumed
Productivity Loss for Inefficient Days	50%	Assumed

Cost Model:

Items	Formula	Value
Daily Cost per Employee	Cost / Working Days	\$454.55
Search Cost	Projects per Year * Daily Cost per Employee * Days Wasted	\$47727.27
Suboptimal Staffing Cost	(Employees * % Suboptimal Staffing) * Daily Cost * (Inefficient Days * Productivity Loss)	\$357954.55
Total Cost	Search Cost + Suboptimal Cost	\$405681.82

Sensitivity Analysis:

		Suboptimal Staffing				
\$405,682		5.00%	10.00%	15.00%	20.00%	25.00%
Employees	550	\$135,227	\$222,727	\$310,227	\$397,727	\$485,227
	650	\$151,136	\$254,545	\$357,955	\$461,364	\$564,773
	750	\$167,045	\$286,364	\$405,682	\$525,000	\$644,318
	850	\$182,955	\$318,182	\$453,409	\$588,636	\$723,864
	950	\$198,864	\$350,000	\$501,136	\$652,273	\$803,409

Low Fidelity Prototype

 Skills Radar

For Coditas

Dashboard

Project Request

Candidate Match

Team Assembly

Skill Update



Username

Job Title

New Request

Project Name

AI Analytics Dashboard

Industry/Domain

Fintech

Project Type

AI / Data Platform

Working in Progress

Project Name

AI Analytics Dashboard

Need

Data Engineer, ML Engineer

Deadline

17/4/2026

Urgency

High

To-do List

• Approve for A,B,C projects

• Review skill and update on projects X finished 12/4/2026

New Recommendations

Projects Name

• Name + role

• Match %

• Key skills matched (Python, ML, Fintech)

• Availability (Available / Busy / % allocation)

• Quick action: "View profile" / "Add to team"

Skills Radar

For Coditas

Dashboard

Project Request

Candidate Match

Team Assembly

Skill Update



Username

Job Title

Upload Transcript or Project Brief

PDF, DOCX, TXT, Transcript

Save Draft

Generate Candidates

Create New Project Staffing Request

Project Name

AI Analytics Dashboard

Industry/Domain

Fintech

Project Type

AI / Data Platform

Project Description

Lorem ipsum color sit amet

Required Skills

Python

SQL

Cloud Engineering

Availability

Role Breakdown

Tech Lead

1

Tech Lead

2

Tech Lead

2

+ Add Role

Start Date

Duration

Timezone / Location

Budget / Rate

Seniority Mix

Junior 20%

Mid 50%

Senior 30%

Client Constraint

CANDIDATE MATCH

Recommended engineers for this request

Review ranked candidates, inspect the evidence behind each recommendation, and build your project team without leaving the comparison flow.

Compare
selected

View selected
(1)

Top Match Match Score 92% Availability 80%

Tony Stark

Senior Data Engineer

Strong fit for fintech analytics work with proven delivery under tight deadlines. [Details](#)

View Profile

Add to Team

Python - Advanced

SQL - Advanced

SAS - Advanced

Top Match Match Score 92% Availability 80%

Tony Stark

Senior Data Engineer

Strong fit for fintech analytics work with proven delivery under tight deadlines. [Details](#)

View Profile

Add to Team

Python - Advanced

SQL - Advanced

SAS - Advanced

RECOMMENDATION DETAILS

Why Tony Stark is recommended

Transparent justification helps project managers understand the recommendation before making staffing decisions.

CANDIDATE SNAPSHOT

Role	Match	Availability
Senior Data Engineer	92%	80%

WHY THIS CANDIDATE FITS?

Strong Python capability for automation and data processing

Prior experience handling financial datasets

Availability aligns with project timeline

EVIDENCE FROM PRIOR WORK

Built data pipelines for Fintech Analytics Platform (2026)

Reduced processing time by 20%

Delivered with minimal supervision under deadline pressure

RISKS TO NOTE

Not fully allocated at 100%, so scheduling still needs confirmation

Skills Radar

For Coditas

- Dashboard >
- Project Request >
- Candidate Match >
- Team Assembly >
- Skill Update >

Username
Job Title



Tony Stark

Senior Data Engineer • 9+years experience

Senior Data Engineer specialising in cloud data platforms, ETL pipelines and analytics systems across FinTech and customer data experience

Data Engineering ETL Pipelines Cloud (AWS) Real-time Data

↑ Sydney, Australia 🏠 Hybrid / remote available

+ Add to Team

👤 Resume

✉ Contact

▼ Availability

Technical Skills

Python

Advanced 7 years

SQL

Advanced 9 years

SAS

Intermediate 4 years

Java

Advanced 7 years

AWS

Advanced 7 years

Domain Experience

FinTech Customer Analytics

Data Platforms

Project Experience

Fintech Analytics Platform (2026)

March 2026 - Present

Senior Data Engineer

Built and maintained data pipelines for a fintech analytics platform.

Python SQL AWS Snowflake

Customer 360 Data Platform (2025)

July 2025 - February 2026

Senior Data Engineer

Developed a centralised customer data platform to unify data from multiple business systems for analytics and reporting.

Python SQL PowerBI Spark

Certifications & Training

→ AWS Certified Data Engineer – Associate

→ Google Cloud Professional Data Engineer

→ Microsoft Fabric Data Engineer Associate (DP-700)

Skills Radar

For Coditas

- Dashboard >
- Project Request >
- Candidate Match >
- Team Assembly >
- Skill Update >

Username
Job Title

Team Assembly Finalisation

📋 Project Details

Selected Team

Complete Team

Tony Stark

Senior Data Engineer

Remove

Python SQL AWS Snowflake

Minh Tran

Data Platform Engineer

Remove

Python SQL Data Engineer

William Smith

Junior Software Engineer

Remove

Python SQL Java

Siva Kumar

Machine Learning Engineer

Remove

Python SQL Machine Learning

Skills Covered

Python ✓
SQL ✓
Snowflake ✓
Java ✓
SAS ✗

Skill Update

Project: Fintech Analytics Program 2026

Engineer: Tony Stark

[View selected team](#)

Detected Skills

Suggested Skill Updates

Skills	Current level	Suggested update	Evidence
Python	Advanced	Advanced	Used in data pipeline
SAS	Intermediate	Advanced	Query optimisation
R studio	Intermediate	Advanced	Built pipeline

Evidence from Project

- Build end-to-end data pipeline
- Optimised SQL queries
- Delivered analytics output

Evidence from Project

- Build end-to-end data pipeline
- Optimised SQL queries
- Delivered analytics output

Confirm or adjust detected skills

[Confirm All](#)

[Edit Skill Level](#)

[Reject](#)

Impact: Confirming this update improve future staff efficiency